Requirements Analysis

DriveAlb Car Marketplace

Prepared for LAB 05 – Requirements Engineering

Project Summary: DriveAlb is a web-based car marketplace where clients browse cars, sellers publish listings, staff verify submissions, and admins manage platform users and activity. This document is organized according to the structure requested in the course slides: User Scenarios List, User Scenarios Extended, Use Cases, Functional Requirements, and Non-Functional Requirements.

1. User Roles

Role	Description
Client	Browses listings, filters cars, views details, sends inquiries, and submits rental requests.
Seller	Creates and manages listings, uploads images, submits listings for verification, and tracks performance.
Staff	Reviews and verifies listings, rejects invalid listings, and manages car brands.
Admin	Manages users, roles, account status, audit logs, and platform-wide activity.

2. User Scenarios List

Scenario ID	User Role	User Scenario
US-01	Client	Register an account
US-02	Client	Verify email after registration
US-03	Client	Log in to the platform
US-04	Client	Recover forgotten password
US-05	Client	Browse available car listings
US-06	Client	Search and filter cars
US-07	Client	View detailed car information
US-08	Client	Send an inquiry about a car
US-09	Client	Submit a rental request
US-10	Client	Update personal profile

Scenario ID	User Role	User Scenario
US-11	Seller	Register as a seller
US-12	Seller	Log in to seller dashboard
US-13	Seller	Create a new car listing
US-14	Seller	Upload images and car information
US-15	Seller	Edit an existing listing
US-16	Seller	Delete an existing listing
US-17	Seller	Submit listing for verification
US-18	Seller	View listing status
US-19	Seller	Re-submit a rejected listing
US-20	Seller	Manage promotions
US-21	Seller	View listing performance and revenue
US-22	Staff	Log in to staff dashboard
US-23	Staff	Review pending listings
US-24	Staff	Approve a listing
US-25	Staff	Reject a listing with a reason
US-26	Staff	Re-review a corrected listing
US-27	Staff	Manage car brands
US-28	Admin	Log in to admin dashboard
US-29	Admin	View all users
US-30	Admin	Manage user accounts
US-31	Admin	Assign or change roles
US-32	Admin	Enable or disable accounts
US-33	Admin	View audit logs
US-34	Admin	Monitor platform-wide activity

3. User Scenarios Extended

Scenario ID	Actor	Goal	Preconditions	Steps	Output
US-01	Client	Create a personal account	User is not logged in	1. Open registration page. 2. Enter name, email, password, and required details. 3. System validates inputs. 4. System creates account in unverified state. 5. System sends verification email. 6. User opens verification link. 7. System marks account as verified.	Verified account is created
US-06	Client	Find a car that matches personal needs	Verified listings exist	1. Open listings page. 2. System displays available cars. 3. User enters keywords or filters such as brand, price, type, or sale/rent status. 4. System updates results. 5. User selects a listing to view details.	Client sees filtered results
US-09	Client	Request to rent a selected car	Client is logged in and car is rentable	1. Log in. 2. Open car details page. 3. Click rent request option. 4. Enter rental details. 5. System validates request. 6. System stores	Rental request is recorded

Scenario ID	Actor	Goal	Preconditions	Steps	Output
				request. 7. System links request to client and listing.	
US-13 / US-17	Seller	Add a new listing and submit it for review	Seller is logged in	1. Open create listing page. 2. Enter car details. 3. Upload images. 4. Select sale or rent mode. 5. Submit listing. 6. System stores listing with 'Pending Verification' status.	Listing is saved and queued for review
US-19	Seller	Correct a rejected listing and send it again	Listing was previously rejected	1. Open dashboard. 2. View rejected listing. 3. Review rejection reason. 4. Edit information or images. 5. Re-submit listing. 6. System changes status to 'Pending Verification'.	Listing returns to review queue
US-23 / US-24 / US-25	Staff	Verify or reject submitted listings	Staff is logged in and pending listings exist	1. Open pending listings page. 2. Select a listing. 3. Review data, images, and rules compliance. 4. Approve listing or reject it with reason. 5. System updates listing	Listing becomes approved or rejected

Scenario ID	Actor	Goal	Preconditions	Steps	Output
				status.	
US-27	Staff	Maintain the list of car brands	Staff is logged in	1. Open brand management. 2. Add, edit, or remove a brand. 3. System validates action. 4. System updates brand data.	Brand catalog is updated
US-30 / US-31 / US-32	Admin	Manage platform users and access	Admin is logged in	1. Open user management. 2. View all users. 3. Select a user. 4. Update role or status. 5. Save changes. 6. System stores the action in audit logs.	User account is updated successfully
US-33	Admin	Review important platform actions	Admin is logged in	1. Open audit log page. 2. System displays recorded actions. 3. Filter or search records. 4. Review results.	Admin can track system activity

4. Use Cases

UC-01 — Register Account

Field	Description
Use Case ID	UC-01
Use Case Name	Register Account
Primary Actor	Client / Seller
Goal	Create a new platform account
Preconditions	User is not logged in

Field	Description
Postconditions	Account is created and marked unverified until email confirmation
Main Flow	1. User opens registration page. 2. User enters required information. 3. User submits the form. 4. System validates the data. 5. System stores the account. 6. System sends email verification link.
Alternative Flows	A1. Email already exists → system shows error. A2. Password is too weak → system shows validation error. A3. Required fields are missing → submission is blocked.

UC-02 — Log In

Field	Description
Use Case ID	UC-02
Use Case Name	Log In
Primary Actor	Client / Seller / Staff / Admin
Goal	Access the system according to role
Preconditions	User has a registered account
Postconditions	User is authenticated and redirected to homepage or dashboard
Main Flow	1. User opens login page. 2. User enters email and password. 3. User submits credentials. 4. System validates them. 5. System creates authenticated session/token. 6. System redirects based on role.
Alternative Flows	A1. Incorrect password → login error. A2. Account disabled → access denied. A3. Email not verified → verification required.

UC-03 — Browse and Filter Cars

Field	Description
Use Case ID	UC-03
Use Case Name	Browse and Filter Cars

Field	Description
Primary Actor	Client
Goal	Find relevant cars from available listings
Preconditions	Approved listings exist
Postconditions	Matching results are displayed
Main Flow	1. Client opens listings page. 2. System shows approved listings. 3. Client enters search or filter criteria. 4. System applies criteria. 5. System shows matching results. 6. Client selects a car to view details.
Alternative Flows	A1. No result matches the criteria → system shows empty-state message.

UC-04 — View Car Details

Field	Description
Use Case ID	UC-04
Use Case Name	View Car Details
Primary Actor	Client
Goal	Read complete information about a car
Preconditions	Listing is approved and visible
Postconditions	Client sees listing details
Main Flow	1. Client selects a car. 2. System retrieves listing information. 3. System displays title, images, brand, model, price, description, and availability.
Alternative Flows	A1. Listing is unavailable or removed → system shows not-found message.

UC-05 — Submit Rental Request

Field	Description
Use Case ID	UC-05
Use Case Name	Submit Rental Request
Primary Actor	Client

Field	Description
Goal	Send a request to rent a car
Preconditions	Client is logged in and listing is rentable
Postconditions	Rental request is stored
Main Flow	1. Client opens car details page. 2. Client clicks rental request button. 3. Client enters required request details. 4. Client submits request. 5. System validates information. 6. System stores request. 7. System notifies seller or records request for review.
Alternative Flows	A1. User is not logged in → redirect to login. A2. Listing is not rentable → request blocked. A3. Required fields missing → validation errors shown.

UC-06 — Create Listing

Field	Description
Use Case ID	UC-06
Use Case Name	Create Listing
Primary Actor	Seller
Goal	Add a new car listing
Preconditions	Seller is logged in
Postconditions	Listing is created with pending verification status
Main Flow	1. Seller opens create listing page. 2. Seller enters car details. 3. Seller uploads images. 4. Seller selects sale or rent mode. 5. Seller submits listing. 6. System validates input. 7. System stores listing as pending verification.
Alternative Flows	A1. Missing fields → submission rejected. A2. Unsupported image format → upload error shown.

UC-07 — Verify Listing

Field	Description
Use Case ID	UC-07
Use Case Name	Verify Listing
Primary Actor	Staff
Goal	Approve or reject a seller listing
Preconditions	Staff is logged in and pending listings exist
Postconditions	Listing is approved or rejected
Main Flow	1. Staff opens pending listings. 2. Staff selects a listing. 3. System displays submitted details. 4. Staff reviews content. 5. Staff approves listing or rejects it with reason. 6. System updates listing status. 7. System records the action.
Alternative Flows	A1. Listing data is incomplete → reject listing. A2. Listing violates platform rules → reject with explanation.

UC-08 — Manage Users

Field	Description
Use Case ID	UC-08
Use Case Name	Manage Users
Primary Actor	Admin
Goal	Control user accounts and roles
Preconditions	Admin is logged in
Postconditions	Selected user information, role, or status is updated
Main Flow	1. Admin opens user management. 2. System displays all users. 3. Admin selects a user. 4. Admin changes role or status. 5. Admin saves changes. 6. System updates the user record. 7. System stores action in audit log.
Alternative Flows	A1. Invalid role assignment → action blocked. A2. Selected user no longer exists → system shows error.

5. Functional Requirements

Requirement ID	Functional Requirement
FR-01	The system shall allow users to register accounts by entering required personal information.
FR-02	The system shall send an email verification link after successful registration.
FR-03	The system shall allow only verified users to log in.
FR-04	The system shall authenticate users with email and password credentials.
FR-05	The system shall support role-based access for Client, Seller, Staff, and Admin users.
FR-06	The system shall allow users to recover forgotten passwords through email-based reset functionality.
FR-07	The system shall display approved car listings to clients.
FR-08	The system shall allow clients to search listings using keywords.
FR-09	The system shall allow clients to filter listings by brand, price, type, and availability.
FR-10	The system shall display detailed information for a selected car listing.
FR-11	The system shall allow clients to submit inquiries for a selected car listing.
FR-12	The system shall allow clients to submit rental requests for rentable vehicles.
FR-13	The system shall allow sellers to create new car listings.
FR-14	The system shall allow sellers to upload one or more images for each listing.
FR-15	The system shall allow sellers to edit their own listings.
FR-16	The system shall allow sellers to delete their own listings.
FR-17	The system shall assign a 'Pending Verification' status to newly submitted or re-submitted listings.

Requirement ID	Functional Requirement
FR-18	The system shall allow staff users to review pending listings.
FR-19	The system shall allow staff users to approve listings.
FR-20	The system shall allow staff users to reject listings and provide a rejection reason.
FR-21	The system shall allow sellers to view the verification status of their listings.
FR-22	The system shall allow sellers to re-submit corrected rejected listings.
FR-23	The system shall allow staff users to manage car brands.
FR-24	The system shall allow admins to view all user accounts.
FR-25	The system shall allow admins to update user roles.
FR-26	The system shall allow admins to enable or disable user accounts.
FR-27	The system shall record critical platform actions in an audit log.
FR-28	The system shall allow admins to view audit logs.
FR-29	The system shall allow sellers to view listing performance or revenue-related information.
FR-30	The system shall allow sellers to manage promotional options for listings.

6. Non-Functional Requirements

Requirement ID	Category	Non-Functional Requirement
NFR-01	Product / User Interface	The user interface shall be accessible through modern web browsers on desktop and mobile devices.
NFR-02	Product / User Interface	The system shall present a consistent interface design across all pages.
NFR-03	Product / Usability	The system shall display validation errors clearly when a user submits

Requirement ID	Category	Non-Functional Requirement
		invalid input.
NFR-04	Product / Performance	The system shall return search and filter results within 3 seconds under normal operating conditions.
NFR-05	Product / Security	The system shall protect passwords using secure hashing algorithms.
NFR-06	Product / Security	The system shall restrict protected resources to authenticated and authorized users only.
NFR-07	Product / Data Handling	The system shall store uploaded images in a secure and organized manner.
NFR-08	Product / Accountability	The system shall preserve audit records for administrative review.
NFR-09	Organizational / Architecture	The system shall be implemented using a separated frontend and backend architecture.
NFR-10	Organizational / Architecture	The backend shall expose structured API endpoints for communication with the frontend.
NFR-11	Organizational / Process	The project source code shall be version-controlled using Git and hosted in a GitHub repository.
NFR-12	Organizational / Maintainability	The system shall support maintainable code organization using modules such as routes, controllers, services, and models.
NFR-13	Organizational / Maintenance	The system shall support basic monitoring and maintenance by logging server-side errors.
NFR-14	Organizational / Deployment	The system shall be deployable in a standard web hosting or server environment.
NFR-15	Availability	The system shall be available to users except during planned maintenance.
NFR-16	Reliability / Security	The system shall prevent

Requirement ID	Category	Non-Functional Requirement
		unauthorized changes to listings, users, and audit records.
NFR-17	Reliability	The system shall validate all required form inputs before data is stored.
NFR-18	Reliability / Error Handling	The system shall handle invalid requests gracefully without exposing internal implementation details.

7. Optional Domain Requirements

Requirement ID	Domain Requirement
DR-01	The system shall distinguish between cars listed for sale and cars listed for rent.
DR-02	The system shall allow rental requests only for listings marked as rentable.
DR-03	The system shall make a listing visible to clients only after staff approval.
DR-04	The system shall associate each listing with exactly one seller account.
DR-05	The system shall maintain car-brand data as a controlled catalog to ensure consistent listing classification.

GitHub-ready note: This document uses the project name DriveAlb throughout and is ready to be added directly to your repository as a deliverable for LAB 05.